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Effects of a Multicomponent Peer Mediated Intervention on Social Communication of Preschoolers With Autism Spectrum Disorder

Ana D. Dueñas[1], Joshua B. Plavnick[2], and Howard Goldstein[3]

Abstract

The absence of direct intervention and peer and staff training can contribute to social isolation in children with autism spectrum disorder (ASD). Effective interventions are needed to realize the benefits of inclusive education. This study examines the effects of a multicomponent, peer-mediated intervention on the social communication behaviors of children with ASD and their typical peers in inclusive preschool classrooms. A multiple-probe design across three dyads was used to examine the effects of the multicomponent intervention on social initiations and social communication exchanges. Improved social communication exchanges between children with ASD and their peers resulted for each dyad. In contrast to previous research, children with ASD demonstrated increased social initiations but less positive effects for social responses. The study contributes to an emerging body of literature that examines the effects of multicomponent peer-mediated interventions on social communication among preschool children with ASD.

Inclusive education for young children with autism spectrum disorder (ASD) is challenging due to the need to balance access to the environment while providing sufficient opportunities to learn at an instructional level that is appropriate for the child with ASD (Lindsay et al., 2013). The social interaction identified as a positive aspect of inclusion can occur only if children with ASD are able to interact with peers and if those peers are willing and prepared to interact with children with ASD (Kasari et al., 2012). In addition, to create an environment in which sustained social interaction can occur, it is important to devise instructional strategies that reduce the involvement of adult interventionists as quickly as possible (Kasari et al., 2012). Procedures that simultaneously teach children with ASD to independently interact with typical peers, and vice versa, have potential to address the need for social inclusion while also delivering targeted social communication interventions.

Early childhood curricula consists of an extensive amount of play to teach social interactions, among other skills, to children. Reciprocal and pretend play sets the stage for social-emotional and language development in young children (Singer et al., 2003). Through play, children experience positive social interactions, such as laughing, verbalizing, smiling, and making eye contact (Gardner & Bergen, 2006). The amount of time that young children spend in these activities depends significantly on their social communication ability (Lin et al., 2019).

Young children with ASD must display socially competent reciprocal play behaviors

[1]Lehigh University
[2]Michigan State University
[3]University of South Florida

Corresponding Author:
Ana D. Dueñas, Lehigh University, 111 Research Drive, Bethlehem, PA 18015, USA.
Email: duenasan@lehigh.edu

with peers to be successful social members of inclusive early childhood classrooms. However, most young children with ASD demonstrate substantial deficits in reciprocal play interactions with peers (Paul, 2003). Therefore, an essential component of an effective approach to inclusion is explicit instruction in play. Many children with ASD may also require additional instruction in prerequisites for successful social play, including joint attention, nonverbal communicative attempts, verbal initiations, and verbal responses (Murdock et al., 2007). Interventions that teach these behaviors as part of inclusive educational settings may improve social communication of children with ASD with peers (Sutton et al., 2019; Whalon et al., 2015).

Young children with ASD must display socially competent reciprocal play behaviors with peers to be successful social members of inclusive early childhood classrooms.

One intervention approach is peer-mediated intervention (PMI), whereby features of the social environment are altered to increase social interactions between children with ASD. In PMI, typically developing peers are taught to initiate social interaction and respond to children with ASD in natural settings (English et al., 1997). PMI is a widely used and socially valid evidence-based intervention for young children with ASD in inclusive educational settings (Goldstein et al., 2014; Whalon et al., 2015; Zagona & Mastergeorge, 2016). Researchers have found that multicomponent PMIs, consisting of both peer coaching and direct intervention to children with ASD, promote substantial gains in total communication acts and lead to reduced isolation in nontreatment settings (Kamps et al., 2014; Kasari et al., 2012; Thiemann-Bourque et al., 2016, 2017). A multicomponent PMI approach combines indirect strategies, such as peer training, with child-direct strategies in order to maximize social communication outcomes of children with ASD (Kamps et al., 2014, 2015; Kasari et al., 2012). The combination of highly effective child-direct strategies with PMI may increase the likelihood that children with ASD improve social communication behaviors and contact naturally occurring consequences, such as peer responding.

Few studies have examined the additive effects of peer training to direct social interventions for children with ASD. One exception is Kasari and colleagues (2012), who investigated the combined effects of direct and indirect social skills intervention strategies. Children with ASD were randomly assigned to treatment groups. One group received direct social skills intervention only, and another received direct social skills intervention with peer mediation. The combined intervention resulted in significant enhancements in classwide peer nominations, teacher ratings of social skills, and reductions in solitary engagement on the playground. However, the specific social-communicative outcomes of children with ASD are unknown, as the observational measures obtained did not specify social engagement at this level.

Thiemann-Bourque and colleagues (2016, 2017) reported promising outcomes in the social communication of children with ASD and their peers using a multicomponent peer mediation approach. Specifically, these investigations have shown that the combined use of child-direct interventions (e.g., alternative and augmentative devices, picture-exchange communication system (PECS; Bondy & Frost, 1998) and PMI improved social communication (e.g., initiations and responses) among young children with ASD and their typically developing peers. In one study, results showed moderate effects for children with ASD in social communication that centered on requesting desired items during snack but revealed that other important social-communicative functions, such as commenting, asking questions, and securing attention during play, were lower (Thiemann-Bourque et al., 2016). Outcomes for social engagement were mixed, with some dyads increasing social engagement only during snack, when the children were requesting from peers, and variable social engagement during sensory and play activities.

Video modeling is a highly effective intervention for teaching children with ASD scripted verbalizations during play (Dueñas

et al., 2019; MacManus et al., 2015), and some of these studies have involved typically developing peers (Dueñas et al., 2019; MacDonald et al., 2009; Maione & Mirenda, 2006). Dueñas and colleagues (2019) found that multiple-exemplar video modeling resulted in improvements in peers' scripted and unscripted social communication during pretend play with children with ASD in an inclusive early childhood setting. As part of this multicomponent PMI, video modeling was presented to both groups of children prior to play sessions as a way to prompt children with ASD to verbalize during pretend play interactions. Although children with ASD increased scripted and unscripted verbalizations toward peers, they differed in their response to the intervention. That is, some participants engaged in higher frequencies of either scripted or unscripted verbalizations. Dueñas et al. hypothesized that peer responses to children with ASD may have influenced this pattern in responding. Whereas children with ASD began to deviate from scripts, their typically developing peer partners were not responding consistently. Typically developing peers did not receive specific training on being responsive to children with ASD other than the video models. Variation exists in both the type (e.g., prompting during intervention) and dosage (e.g., before, during, and after intervention) of typical peer training in PMIs, which may lead to variation in the responsiveness of typical peers (Osborne et al., 2019).

Peer-training strategies that reduce adult reliance (e.g., prompting during interactions) are needed to promote maintenance of intervention effects. Self-management is one strategy that is used extensively to promote independent use of taught behavior. *Self-management* is defined as the self-application of some or all components of behavior change strategies that result in desired behavior (Mace & Kratochwill, 1988). Sainato and colleagues (1992) and Osborne et al. (2019) used self-management strategies to teach typically developing preschoolers to self-evaluate strategies taught during PMI (e.g., staying in proximity, getting friend's attention, getting friend to play, sharing with their friend, and talking to their friend). Sainato et al. found that social behaviors of participants toward children with ASD increased when teacher-delivered prompts decreased, whereas Osborne et al. found that a combination of strategies, self-management and recorded reminders were most effective. Self-management is a strategy that may teach typically developing preschoolers to respond reliably to the social initiations of children with ASD across contexts and when children deviate from scripts.

Peer-training strategies that reduce adult reliance (e.g., prompting during interactions) are needed to promote maintenance of intervention effects.

To achieve reciprocal social communication, researchers have used multicomponent interventions to teach increased initiations and responses by both groups of children (Gunning et al., 2018). Despite positive outcomes in some social contexts, there are aspects of multicomponent interventions that have yet to be investigated. Many researchers include adult support within the interventions, and few studies incorporate self-management strategies for peer partners to promote independence on the part of typical peers. Additionally, most studies investigate the extent to which children with ASD respond to peer initiations, with limited research on initiations other than verbal requests by children with ASD.

The purpose of the current study was to examine a multicomponent PMI to teach typically developing children to independently respond to children with ASD during pretend play activities and multi-exemplar video modeling to promote unscripted initiations of children with ASD (see Dueñas et al., 2019). We asked the following research questions:

1. Is there a functional relation between the multicomponent PMI (i.e., video modeling and peer self-management) on social communication exchanges of typically developing preschoolers and children with ASD?
2. What are the generalization effects of the multicomponent PMI on social communication across play activities?
3. What is the frequency of unscripted verbal social initiations and responses of children with ASD?

4. Is the multicomponent PMI acceptable to typically developing preschoolers, and do social communication exchanges improve as compared with normative samples of typically developing peer interactions?

Method

Participants

Participants were recruited from a community-based early intensive behavior intervention (EIBI) program affiliated with a university. Three students with ASD who received inclusive educational programming (Wilmer, Olin, and Titus) and three typically developing peers (Malik, Aria, and Olga) participated. Inclusion criteria for children with ASD were (a) a confirmed diagnosis of ASD by a licensed clinical psychologist, (b) ability to imitate a vocal model, (c) ability to attend to a video for 20 s, (d) ability to play functionally with toys, (d) use of two- to three-word phrases to communicate, (e) limited peer interaction skills as demonstrated by observations in the inclusive classroom during unstructured play, and (f) a minimum of 2 hr of instruction per day in the inclusive classroom including unstructured time that allowed children to choose their play activities. The children with ASD were paired with three typically developing students to form dyads for the duration of the study. Dyadic pairings were decided by the board-certified behavior analyst and the general education teacher, based on their knowledge and observations of the children in the classroom.

Typically developing peers were nominated for this study by their teachers as meeting the following criteria based on previous PMI research (Gunning et al., 2018): (a) being well-liked by typical peers, (b) being a positive social partner with the focal child, (c) being generally compliant with adult directives, (d) ability to attend to a task or activity for at least 10 min, (e) willingness to participate, and (f) ability to attend school on a regular basis. Information regarding typically developing peers was gathered during informal interviews with their teachers.

Wilmer was 4 years 8 months at the time of the study. He spent approximately 5 hr in the EIBI classroom and 3 hr per day in an inclusive preschool classroom for typically developing 4-year-old children. Classroom observations of Wilmer revealed that he joined peers during routine activities (e.g., snack, structured play, art) but did not initiate or respond to peer invitations to play or bids for attention. Wilmer was observed sustaining parallel play with peers but not initiating interactions with peers during play.

Malik was a 4-year-old boy who had attended the preschool classroom 45 hr per week for approximately 1 year. He demonstrated age-appropriate social skills, language and communication, and play skills. He followed adult directives and often volunteered to assist in the classroom. His teacher described him as creative, outgoing, and inquisitive.

Olin was 4 years 3 months at the time of the study. He spent approximately 4 hr in the EIBI classroom and 4 hr per day in an inclusive preschool classroom for typically developing 4-year-old children. Observations revealed that Olin joined peers during preferred activities (e.g., snack, car, and truck toys) but responded inconsistently to peers' social initiations. Olin did not engage in sustained play with a peer.

Olga was a 5-year-old girl who had attended the preschool classroom 45 hr per week for approximately 1 year. She demonstrated age-appropriate social, language and communication, and play skills. She followed adult directives and often volunteered to assist in the classroom. Her teacher described her as outgoing, sometimes bossy, and always willing to help other students.

When pairings were decided, Titus's inclusion teacher was out ill and the teaching assistant nominated two different typical peers, Aria and Genevieve. During initial baseline, Titus was tentatively paired with both typical peers. After the fifth baseline session, Titus was subsequentially paired with Genevieve. After Baseline Session 9 and during peer training, Genevieve refused to participate in the study. Baseline was reinstated with Aria,

and Titus was switched to Aria for the remainder of the study.

Titus was 3 years 7 months at the time of the study. He spent approximately 6 hr in the EIBI classroom and 2 hr per day in an inclusive classroom for typically developing 4-year-old children. Observations revealed that Titus did not join peers during any activities and instead would roam the room and engage in solitary play. Titus did not engage in sustained play with peers and was not responsive to initiations by peers.

Aria was a 3-year-old girl who had attended the preschool classroom 40 hr per week for 6 months. She demonstrated age-appropriate social, language and communication, and play skills. She followed adult directives and often volunteered to assist in the classroom. Her teacher described her as shy and soft-spoken.

Genevieve was a 4-year-old girl who had attended the preschool classroom 45 hr per week for approximately 1 year. She demonstrated age-appropriate social, language and communication, and play skills. She followed adult directives but was sometimes inconsistent. Her teacher described her as shy, sometimes stubborn, and creative.

Setting

The participant dyads received services in inclusive early childhood classrooms located in three different buildings that housed the EIBI programs; one of the early childhood centers had six different early childhood programs, including the EIBI program. The other was housed within a university-based child development program, and the third was housed within a Head Start program. The inclusive classrooms had a 1:8 adult-to-student ratio and were organized in stations that included a section for large-group instruction, four tables for small-group instruction, a reading area, a pretend play area, and an art area. All study sessions were conducted in the pretend play area of the classrooms, and peer-training sessions were conducted in a quiet room adjacent to the children's classroom. However, Titus and Aria's sessions were conducted in an unoccupied classroom adjacent to the children's classroom upon the request of the lead teacher due to other children in the classroom constantly requesting to be part of the intervention.

Materials

A total of three play sets were used during play sessions between the children. Various play sets were selected to promote generalization across toys for children with ASD. The play sets included a wooden tree house play set from Imaginarium, a wooden rocket ship play set from Hape, and a wooden castle from Melissa & Doug. The play sets were 40 to 60 cm tall and contained eight to 10 smaller pieces. Researchers used a GoPro Hero4 action camera to film sessions for later coding and for use during self-management training sessions with peers. An Apple iPad was used to view videos with an application called My Pictures Talk. Children also used headphones to listen to the video models. For peer training, a self-management checklist, token board, identified reinforcers, marker, and laptop were used to view videos. For video coding, the first author created an Excel spreadsheet and used video storage software.

A total of six videos were created, two videos for each of the three play sets. Multiple video exemplars that modeled variations of social initiations were used to increase the likelihood of novel social communication among children with ASD. The 30-s videos modeled 10 social initiations with the play set. Only the toy figurines and adult hands holding the figures were shown in the video, and the same male and female adults modeled the audio scripts. Social initiations modeled were two- to three-word utterances derived from observations of typically developing children in the early childhood classroom (see Dueñas et al., 2019). The video clips showed adults modeling a range of social initiations: (a) request for another to perform an action (e.g., "Dance, Bunny!"), (b) requests for an object (e.g., "Let's trade"), and (c) a vocal label of one's own actions that occasioned a response (e.g., "I'm going to sleep"). The videos varied the modeled play actions, modeled

verbalizations, and stimuli (e.g., toy fox versus toy bunny).

Dependent Measures

A trained graduate student served as independent observer and coded dependent variables from the 4-min videos of play sessions. A total of six variables were coded using a Microsoft Excel spreadsheet created by the first author. Each time a social behavior was observed by each of the participants, the time stamp was entered in a spreadsheet in the appropriate column and "1" was marked on each corresponding column for the dependent variables observed (i.e., social responses, social initiations, social communication exchanges, and unscripted verbalizations). The primary dependent variable was the frequency of social communication exchanges. The secondary dependent variable was the frequency of social initiations and responses by children with ASD. The frequency of unscripted verbalizations by children with ASD was also counted.

Social Communication Exchanges. Social communication exchanges were defined as independent verbal exchanges among children with ASD and their peers. A verbal communication exchange involved a verbal social initiation from either child and a matching verbal response from the communication partner (social initiation + social response) occurring within 5 s of the initiation.

Social Initiations. Social initiations were defined as independent vocal statements that were either a request for another to perform an action (e.g., "Push me, Bunny!"), requests for an object (e.g., "I want the owl"), or a vocal description of one's own actions that occasioned a response (e.g., "Bunny, I'm falling"). Initiations were scored each time a participant emitted a statement that was not explicitly related to a prior statement made by a social partner. For example, a child saying, "Help, I'm falling," would be scored as an initiation. The peer responding, "OK, I'll catch you," followed by "Let's dance!" would be scored as a response and then another initiation for the second part of the statement. Social initiations were not counted if they occurred following a video model presentation during the play session (see Procedures).

Social Responses. Social responses to social initiations were defined as vocal statements that occurred within 5 s following a vocal social initiation and had to explicitly react to the content or context of the preceding social initiation. These could be answers to questions (e.g., "What do you want to eat?" "I want pizza!"), comments or acknowledgments (e.g., "Going on the swing!" "OK, me, too."), or offering assistance (e.g., "Help, I'm falling"; "I'll help you, Bunny"). Responses could be accompanied by movements or gestures (e.g., pointing, tapping on shoulder, actions with toys as agents).

Unscripted Verbalizations. Any verbal social initiation made by the child with ASD that did not match statements modeled in the video by more than conjunctions, articles, prepositions, or pronouns was counted as an unscripted social initiation. In addition, any response by the child with ASD to the typically developing peer that was not directly modeled in a video was counted as an unscripted verbalization. For example, "I want to be the bunny" was counted as an unscripted social initiation and "OK, I will be the fox" as an unscripted response, as these were not modeled in the video.

Interobserver Agreement

The first author trained two graduate students to code interobserver agreement (IOA), who were blind to the intervention, using sample videos of play sessions from a different study. Multiple training videos were used until 90% IOA was achieved between coders. This took on average two to three videos. Agreement was achieved when both coders agreed on whether a social communication exchange had occurred and the type of verbal social communication (e.g., social initiation, social response). When disagreements occurred

during training, these were discussed with the first author to reach consensus. Point-by point agreement was calculated by dividing the number of agreements by the total number of disagreements plus agreements to yield a percentage. IOA was calculated for 30% of randomly selected videos across conditions and participants. Mean IOA between the two coders during baseline and generalization for social communication exchanges was 100%, for social initiations was 100%, for social responses was 100%, and for unscripted verbalizations was 100%, across all participants. Mean IOA during intervention and maintenance for social communication exchanges was 84% (range: 63–100), for social initiations was 87% (range: 60–100), for social responses was 87% (range: 50–100), and for unscripted verbalizations was 75% (range: 50–100), across all participants. When disagreements occurred among the two independent coders, these were resolved by the first author by recoding until agreement was achieved with one of the coders.

Experimental Design

A multiple-probe design (Ledford & Gast, 2018) across dyads was used to assess the effects of the multicomponent intervention on social communication exchanges between children with ASD and their peers. Single-case experimental design standards established by What Works Clearinghouse (WWC) were followed and met for this design by systematically manipulating the independent variable, measuring the outcome variable(s) repeatedly over time, collecting IOA for 30% of sessions across conditions, meeting at least 80% agreement for the primary dependent variable (i.e., social communication exchanges), demonstrating intervention effects across three participants, and collecting at least three data points per phase (WWC, 2017). The primary dependent variable, frequency of social communication exchanges, was used to evaluate the effects of the intervention and make decisions regarding the introduction of treatment to subsequent dyads. Specifically, when a 50%-or-above increase in social communication exchanges was observed over two sessions, the subsequent dyad was introduced to intervention.

Procedures

Two 4-min play sessions were delivered daily, four times per week, and occurred 20 min apart during a 1-hr child-choice period. The 4-min play sessions allowed sufficient data collection while providing children ample time to choose activities and play partners during the child-choice period. The intervention was delivered for 15 sessions for all participants to assess the magnitude of change given a consistent and relatively brief dosage that may be feasible for practitioners. Fifteen sessions were the equivalent of a 5-week intervention for a specific targeted skill, which is a similar time frame used in other social skills interventions (Kamps et al., 2015).

Initial Peer Training. The researcher conducted a one-time, 20-min training with typically developing peers. The training involved (a) typically developing peer assent to intervention by writing their name on a document that stated, “I [name of typically developing peer] agree to be a friendship leader for [name of child with ASD] by helping him/her play, take turns, and use words”; (b) reading a children’s book titled *Pete the Cat and the New Guy* (Dean & Dean, 2014) to introduce the value of playing with those who may behave differently or have different abilities; and (c) showing a 30-s video clip that showed the rules for playing with your friend: eyes on your friend, stay close to your friend, listening ears on, and answer fast, adapted from Stay, Play, Talk (English et al., 1997).

Baseline. During baseline, the typically developing peer invited the child with ASD to play and led them to the play set previously set up on the floor of the pretend play area. The children were instructed to play together and were given 4 min while the researcher remained 1 m away. If children with ASD attempted to leave the play area, the researcher walked over and physically prompted them back to

the play set. If typically developing children attempted to leave the area or became distracted by others, they were verbally prompted to continue playing.

Intervention. During intervention, the same procedures as baseline were followed except self-management was delivered to the typically developing peers 15 min prior to the first play session of the day, and the child with ASD was shown a video model immediately before every play session.

Self-management. A self-management system was modeled from *Self & Match* by Salter and Croce (2013). The typically developing children were taught to self-monitor the extent to which they responded appropriately to initiations of their peers with ASD from a video of the previous play session. For the first self-management intervention, a video was shown from the final baseline play session. For subsequent self-management intervention sessions, a video was shown from the previous intervention session. The behaviors that were self-monitored were (a) looking in the direction of the child with ASD when the child socially initiated (eyes on your friend), (b) staying within arm's length distance (stay close to your friend), (c) listening to their friend (listening ears on), and (d) responding to their friend within 5 s of a social initiation (answer fast).

For self-management, the researcher brought the typically developing peer into a quiet room adjacent to their classroom and explained that they were going to view themselves playing with their friend and see if they followed the rules for playing with their friend. Second, the researcher restated the rules (i.e., eyes on your friend, stay close to your friend, listening ears on, and answer fast) and explained they were going to play a game called Match for Yes! Third, the researcher explained the rules of the Match for Yes! Game: The peer would color a happy face if they saw themselves performing each behavior or would color a sad face if they did not see themselves performing the behavior from the video, and the researcher would do the same. If they matched for yes with the researcher, they would get 1 point; if they matched for no, they would get half a point; and if they did not match, they would not get any points. Points were exchanged for reinforcers that were assessed prior to the start of the study. Fourth, the researcher modeled the use of the self-management checklist with a video model from a baseline session for practice (this happened during the first self-management training session only).

Once the child received two opportunities for practice, two video segments were selected from the previous 4-min play session, one following the first initiation by the child with ASD and one following the last initiation from the child with ASD. The researcher showed the 5- to 7-s video segment, paused the video to ask if the child observed themselves performing the behavior or did not observe themselves perform the behavior, and asked the child to color in yes or no (i.e., happy or sad face); the researcher also colored yes or no, based on what was observed. This was repeated for each behavior on the self-management checklist. As the listening-ears behavior was not observable, this was given as a free point to ensure typically developing peers received reinforcement during the acquisition phase. Once all videos were shown, the number of matched happy faces were counted, and points were totaled and tracked across sessions. Once a child earned 20 points, they exchanged points for a terminal reinforcer. Following self-management, children returned to the classroom and play sessions were conducted.

Video-modeling presentation. Children with ASD were presented with video models corresponding to each play set. Video exemplars were rotated across sessions to ensure each play set and video was presented five times across 15 sessions. The researcher presented a video model to the child with ASD that corresponded to the play set as a procedure to prompt social initiations immediately before each play session and during play sessions, if needed (see Play Sessions).

Play sessions. After the typically developing child received self-management and immediately following video modeling for children with ASD, a play session was conducted. During play sessions, the typically developing child was told to set up the toys while their friend watched the video. Once the child viewed the video, the typically developing child was told to invite their friend to play. As in baseline, the children were given 4 min to play, while the researcher remained 1 m away. If the child with ASD did not verbally initiate after 30 s, the researcher walked over to the child and said, "Play like you saw in the video." If after this initial verbal prompt the child with ASD still did not imitate the verbal social initiations modeled in the video, the 4-min play session was paused, and the researcher showed the child a randomly selected 5-s clip from the video model on the iPad and said, "Play like you saw in the video." No additional prompts were delivered to either child.

Maintenance. Three maintenance probes were conducted 3 weeks after children completed their 15th intervention session. The same conditions as in baseline were presented during maintenance probes to assess whether treatment gains persisted beyond intervention sessions and in the absence of video models and peer training.

Stimulus Generalization. Generalization probes were conducted once during baseline, after every four or five sessions of intervention, and once after maintenance to assess stimulus generalization. Generalization probes were conducted using play sets that were available in the preschool classroom and varied for each dyad (e.g., pretend veterinary kit, Lego people, and baby dolls). As in baseline, the children were instructed to play together for 4 min. No further prompting was delivered. Generalization probes were conducted to assess whether participants with ASD and typically developing peers generalized social communication behaviors with materials dissimilar to intervention.

Procedural Fidelity

As shown in Table 1, checklists were created to capture the steps required for implementation of the direct intervention (i.e., multiple-exemplar video modeling) and the steps for implementation of the peer training (e.g., self-management and token economy). A behavior technician in the inclusive classroom, blind to the purpose of the study, was trained to use the checklists and completed procedural integrity checks in vivo on 30% of randomly selected sessions across conditions and participants. The average procedural integrity during baseline and generalization across participants was 100%, during typical peer intervention was 100%, and during play sessions across all dyads was an average of 95% (range: 82–100).

Social Validity

Social validity was assessed by obtaining descriptive feedback from typical peers and comparing the play sessions with those of typically developing peers. Teaching assistants administered a 6-item questionnaire to typical peers (cf. Table 2). During the interview, the teaching assistant sat across from the peers and asked them to circle responses after reading each item. Responses were measured on a 5-point Likert-scale illustrated by happy-to-sad-face gradations. The peers responded to the questionnaire at three time points: after three sessions of intervention, after the ninth intervention session, and after the 3-week follow-up.

We also evaluated the extent to which the social communication exchanges of participants with ASD and typically developing peers matched those of typically developing peers. The teacher in the inclusive classroom nominated an additional peer in the classroom for this comparison. The two typically developing peers played for five 4-min play sessions using the same intervention play sets. Procedures were identical to baseline. The frequency of social communication exchanges averaged across the three dyads served as a normative comparison.

Table 1. Procedural Integrity Form (Intervention).

Date	Session	Initials	Data	
Before play session (video modeling)				
1. Asks typical peer to invite their friend to play after they see a video			Y	N
2. Asks child with ASD to come watch a video			Y	N
3. Shows correct video model (video is rotated, matches play set)			Y	N
4. Shows video model twice			Y	N
5. Ensures child is watching video			Y	N
6. Says, "Go play like the video" after peer invites child to play			Y	N
During play session				
7. Remains 3 feet away			Y	N
8. Does not deliver any prompts during play			Y	N
9. If 30 s lull in interaction, represents 5 s of video clip and says, "Play like the video"			Y	N
10. If children leave the play area, prompts them back to play area			Y	N
11. Ends play session after 4 min			Y	N
SUBTOTAL				
Typical peer self-monitoring				/11
1. Reviews rules for being a good play partner			Y	N
2. Tells child that they are going to watch a video of themselves playing			Y	N
3. Plays video clip of first social initiation by child with ASD			Y	N
4. Asks child whether they performed each behavior			Y	N
5. Notes whether child performed the behavior			Y	N
6. Goes through each behavior and notes matches and states points received			Y	N
7. Asks child to count points			Y	N
8. Plays video clip of last social initiation by child with ASD			Y	N
9. Asks child whether they performed each behavior			Y	N
10. Notes whether child performed the behavior			Y	N
11. Goes through each behavior and notes matches and states points received			Y	N
12. Asks child to count points			Y	N
13. Transfers points to point chart			Y	N
TOTAL				/24

Note. ASD = autism spectrum disorder.

Results

Social Communication Exchanges

Figure 1 shows the frequency of independent social communication exchanges across dyads during baseline, intervention, maintenance, and generalization probes. Following the stable baselines, we observed a functional relation between the multicomponent PMI and frequency of social communication exchanges that varied across dyads.

Social communication exchanges between Wilmer and Malik were stable during baseline and averaged approximately one exchange per play session. Effects of the multicomponent intervention were not immediate. During intervention, we observed an increasing trend of social exchanges, ranging from none to 13 and two overlapping points with baseline. On average, four social communication exchanges were observed during intervention that maintained at an average of five social communication exchanges per play session at 3 weeks postintervention with minimal variation. Social communication exchanges also generalized to a different set of toys in their

Table 2. Typical Peer Survey Results.

	Typically developing peers								
	Malik			Olga			Aria		
Item	Beginning	Middle	End	Beginning	Middle	End	Beginning	Middle	End
How much do you like helping [peer with ASD]?	1	1	1	1	5	5	1	1	3
How much do you like watching the videos?	1	1	1	1	1	1	1	1	1
How much fun do you have playing with [peer with ASD]?	1	1	3	1	3	3	1	1	2
Would you help [peer with ASD] play again?	1	1	3	1	5	4	1	1	1
How hard was it to play with [peer with ASD]?	5	1	1	5	5	4	5	5	3
How much did you like leaving your classroom?	1	1	1	1	4	1	1	1	1

Note. ASD = autism spectrum disorder; 1 = a lot/very much will; 2 = liked it/yes; 3 = somewhat/maybe; 4 = a little/probably won't; 5 = not at all/definitely won't.

classroom, from once at baseline to an average of four exchanges during generalization (range: 0–6).

We observed a slight increasing trend in social communication exchanges between Olin and Olga followed by a decreasing trend during baseline; on average, three exchanges occurred per play session. However, the effect of the multicomponent intervention on social communication exchanges was immediate for this dyad, and the level of social communication exchanges was significantly above baseline with one overlapping data point between baseline and intervention. Social exchanges were variable, following an increasing trend followed by a slight decrease and subsequent increasing pattern. An average of 13 social communication exchanges (range: 7–22) were observed that maintained at an average of 17 per play session at 3 weeks post intervention. Social communication exchanges did not generalize to a different set of toys in their classroom. Eight exchanges were observed at baseline, and an average of seven exchanges were observed during intervention (range: 0–9).

We observed a stable baseline pattern for Titus, who displayed an absence of social communication exchanges with both Aria and Genevieve. Baseline Sessions 1 through 5 represent data with Aria, Sessions 6 and 7 represent data with Genevieve, and Session 8 and 9 represent data with Aria. We observed no overlap between baseline and intervention. Effects of the multicomponent intervention were immediate for Titus and Aria. During intervention, social exchanges were somewhat variable: We observed an increasing trend, followed by a decreasing trend and an increase toward the end. The overall level was significantly above baseline and averaged 10 social communication exchanges (range: 4–16). Intervention levels maintained at an average of nine per play session at 3 weeks postintervention. Social communication exchanges generalized to a different set of toys in their classroom, although the average frequency during this condition was much lower than intervention, from zero at baseline to two exchanges per play session (range: 0–5).

Social Initiations and Responses of Children With ASD

Figure 2 shows the frequency of independent social initiations and responses by children with ASD during baseline, intervention, maintenance, and generalization probes. We

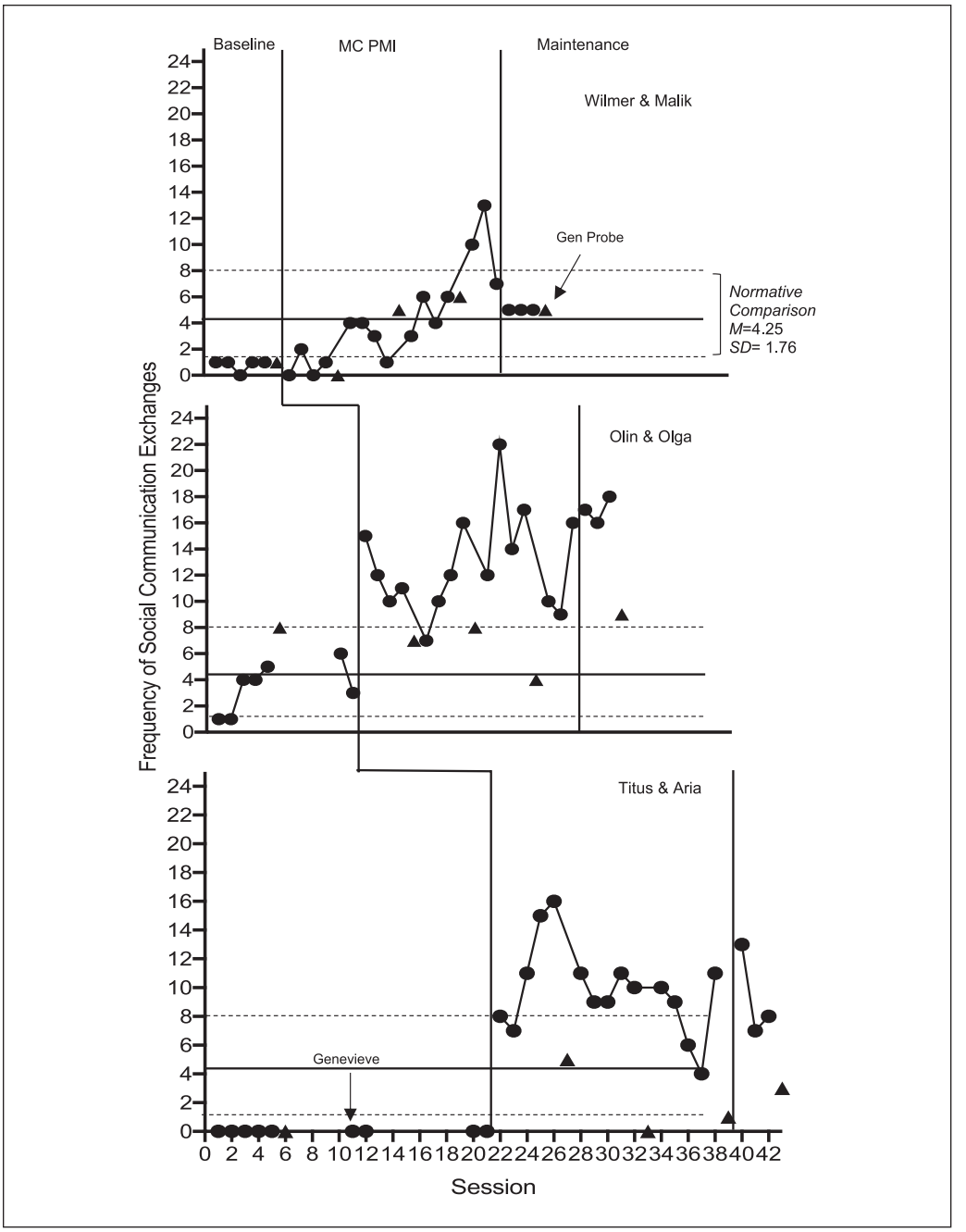


Figure 1. The frequency of independent social communication exchanges across dyads is shown during baseline, intervention, and maintenance in closed circles. Generalization probes during baseline, intervention, and maintenance are depicted in closed triangles. Means and standard deviations of social communication exchanges across 12 observations and three dyads are shown on the solid vertical line and dotted lines to represent the upper and lower bounds for normative comparison. MC PMI = multicomponent peer-mediated intervention.

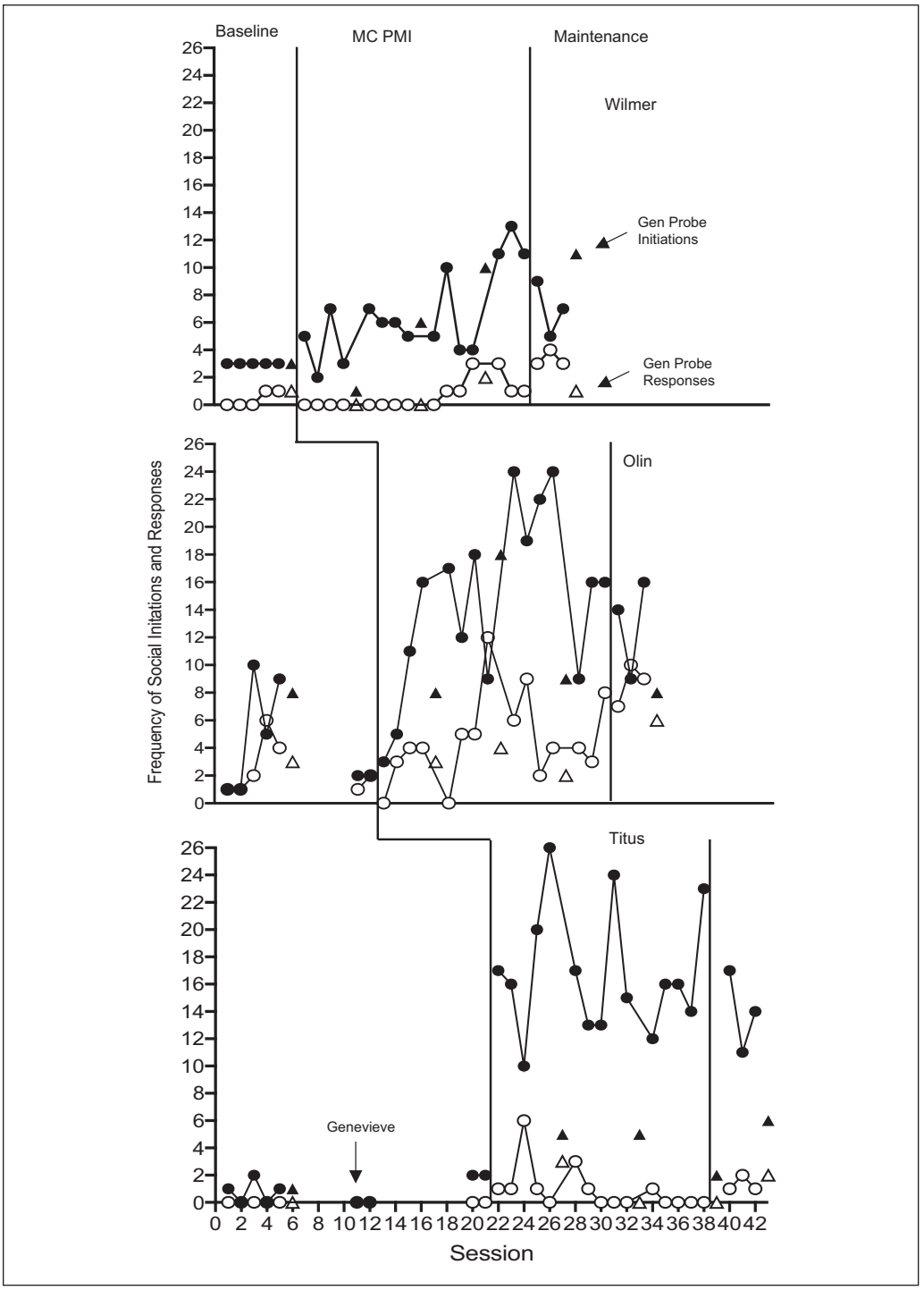


Figure 2. The frequency of independent social initiations is shown in closed circles, and social responses are shown in open circles across baseline, intervention, and maintenance by children with ASD. Generalization probes are depicted in closed triangles for social initiations and open triangles for responses across baseline, intervention, and maintenance. MC PMI = multicomponent peer mediated intervention.

observed a functional relation between the multicomponent PMI and social initiations of children with ASD but did not observe a functional relation for social responses.

We observed stable baseline for Wilmer's social initiations, averaging three per play session. There is one overlapping point across baseline and intervention and one point below baseline levels in intervention. However, we observed an immediate change in trend, with a slow increasing pattern and some variability in social initiations. Overall, Wilmer's level of initiations was higher than baseline and on average occurred seven times per play session (range: 2–13). Social initiations maintained at seven (range: 5–9) 3 weeks postintervention. Wilmer's social initiations to his peer also generalized to a new set of toys from the inclusive classroom. His average social initiations during generalization probes improved to seven from only three at baseline.

Baseline for social responses were stable, as Wilmer was not responding to his typically developing peer. There is significant overlap in data between baseline and intervention. We observed no change in the effects of the intervention on social responses and a minimal increase toward the end of the intervention. During maintenance probes, Wilmer maintained a slightly higher frequency of social responses at 3 weeks postintervention, an average of three per play session. Social responses did not generalize to a new set of toys.

Baseline levels were low with some variability for social initiations for Olin; on average, four social initiations per play session (range: 1–10) were observed. There is overlap in data between baseline and intervention. However, an increasing trend in social initiations is clearly observed with some variability and an increase in level of initiations. Social initiations averaged 15 per play session (range: 3–24). Effects of the intervention were not immediate. Social initiations maintained at an average frequency of 13 (range: 9–16) 3 weeks postintervention. Olin's social initiations to his peer also generalized to a new set of toys from the inclusive classroom. Average social initiations during generalization probes increased to 11 from eight during baseline.

We observed low levels of responding with some variability for Olin during baseline. There is significant overlap in data between baseline and intervention. However, we observed a change in trend and level of responses, but these increased only to an average of four responses per play session (range: 0–12). Effects of the intervention for social responses were minimal with high variability. During maintenance probes, Olin responded slightly more consistently at an average of nine times (range: 7–10). However, these did not generalize to a new set of toys in the inclusive classroom. During baseline generalization probes, he responded three times and he continued to respond an average of three times during intervention.

We observed stable baselines for Titus, who initiated to his peer zero to two times during play sessions and no overlap in data between baseline and intervention. During intervention, we observed an immediate effect of the intervention, averaging 17 times per play session (range: 10–23) with some variability across play sessions. Social initiations maintained at an average frequency of 14 three weeks postintervention. Titus's social initiations to his peer also generalized to a new set of toys from the inclusive classroom. However, average social initiations during generalization probes were lower than intervention at five per play session from zero during baseline generalization probe (range: 2–6).

Stable baseline patterns were observed for Titus, who did not respond to his peer, as was significant overlap in data between baseline and intervention. During intervention, responses remained low at an average of one response per play session (range: 0–6). There were no effects of the intervention on social responses. During maintenance and generalization probes, Titus responded at similar levels (range: 0–3).

Unscripted Verbalizations

We also were interested in whether children with ASD began to deviate from scripts across play sessions. Figure 3 shows the frequency

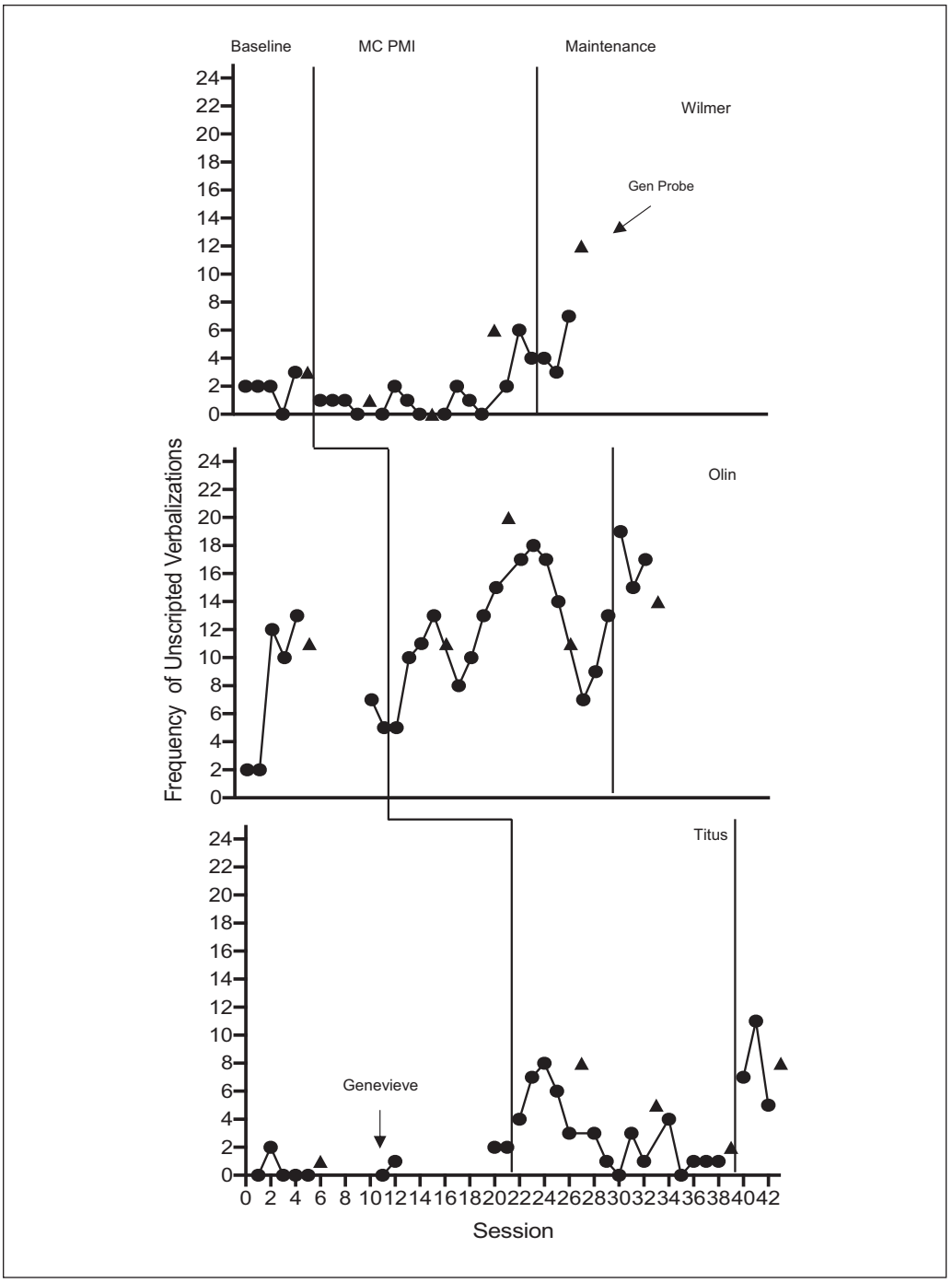


Figure 3. The frequency of unscripted social initiations and responses is shown in closed circles during baseline, intervention, and maintenance across participants with ASD. Generalization probes during baseline, intervention, and maintenance are depicted in closed triangles. MC PMI = multicomponent peer mediated intervention.

of unscripted verbalization during baseline, intervention, maintenance, and generalization probes across participants. In summary, there was no observable change in level or trend from baseline to intervention across dyads. Thus, a functional relation was not observed between intervention and unscripted verbalizations.

Tau-U Effect Sizes

To supplement visual analysis, we calculated a Tau-U effect size metric for the main dependent variables of interest (i.e., social communication exchanges, social initiations and responses). Tau-U offers nonparametric effect sizes by calculating the nonoverlap of data between baseline and intervention phases while controlling for possible trends during baseline. The effect size can be interpreted as small (0.65 or lower), medium to large (0.66–0.92), and large (0.93–1.0) (Parker & Vannest 2009; Rakap, 2015). A web-based Tau-U calculator (see http://www.singlecaseresearch.org/calculators/tau-u) was used for calculations. Weighted Tau-U for social communication exchanges across dyads was .89, 95% confidence interval [CI] [0.58, 1], $p < .001$; the weighted Tau-U for social initiations of children with ASD was .88, 95% CI [0.59, 1], $p < .001$; and the weighted Tau-U for social responses was .36, 95% CI [0.05, 0.67], $p < .02$. These metrics were consistent with visual analysis.

Typically Developing Peer Survey Results

Typically developing peers rated the extent to which they enjoyed different features of participation in the study (e.g., leaving their classroom for training, watching videos, and playing with a child with ASD) and on whether they would like to participate in the future. See Table 2 for a summary of responses for surveys conducted at the beginning, middle, and end of the study. Overall, typically developing peers rated the intervention positively at the beginning of the study, but two participants rated the intervention less favorably toward the end of the study. When the teaching assistant asked the children why they felt this way, both children stated that their play partner with ASD was not sharing toys.

Normative Comparison Results

The second form of social validity assessed the clinical significance of our primary dependent variable of interest, social communication exchanges, by comparing baseline and intervention levels of participant dyads with those of typically developing peer dyads in the same classroom. Typically developing peer comparison data are shown in Figure 1. Based on 12 total observations across three typically developing peer dyads, typical peers' frequency of social communication exchanges had a mean of 4.25 ($SD = 1.76$, $SE = 0.509$, 95% CI [3.25, 5.24]). For two of three participant dyads, the average social communication exchanges observed during baseline were consistently below the normative range. During intervention and maintenance, the frequency of social exchanges exceeded or greatly exceeded the normative range.

Discussion

The multicomponent PMI led to an increase in social communication exchanges between children with ASD and typically developing peers with minimal adult mediation during play sessions. Children with ASD also increased social initiations directed to their typically developing peers. In addition, two dyads generalized social communication exchanges to toys in the inclusive classroom. However, experimental effects were modest for social responses of children with ASD, and the observed effect for response generalization (unscripted verbalizations) was minimal.

The current PMI intervention literature has documented that teaching children with ASD to socially initiate can be especially difficult during play (Dotson et al., 2010; Thiemann-Bourque et al., 2017). In the present investigation, we successfully taught children with ASD to initiate verbal play statements. Video modeling shows promise as part of a multicomponent PMI for teaching children with

ASD to initiate to peers. This study expands the use of video modeling to promote social initiations of children with ASD in Dueñas et al. (2019) by specifically teaching typically developing peers to respond to children with ASD. As a whole, the self-management and video-modeling package shows promise as an intervention to promote verbal social communication exchanges during play. In addition, the intervention was conducted in only 15 play sessions that spanned approximately 2 weeks, and effects were maintained 3 weeks postintervention.

Although the desired effect of the intervention on social communication exchanges was achieved, an unintended effect was that children with ASD were primarily initiating and being responded to. This has important implications for the design and measurement of future PMI studies. It seems necessary to teach both groups of children to respond and initiate. Although social communication exchanges as a measure of reciprocity may capture some degree of balance between interactions, the direction of this interaction (i.e., who responds and who initiates) may also be essential to achieve reciprocity. Unlike previous studies, we focused the child-direct intervention on teaching children with ASD to initiate to their peers during play. Therefore, our findings differed from Thiemann-Bourque et al.'s (2016, 2017) results, in which social initiations by children with ASD occurred at lower rates and initiations of typical peers occurred at higher rates. Notably, our intervention focused on teaching diverse forms of social communication, other than requesting desired items.

Generalization across stimuli was observed for social communication exchanges for two dyads, and unscripted social initiations were observed for two children with ASD. Although generalization is measured infrequently in PMI (Goldstein et al., 2014; Gunning et al., 2018), our findings are similar to recent studies that have found generalization effects to be mixed within the PMI literature (Whalon et al., 2015).

The stimuli selected for generalization probes within this study may have affected the results of generalization. We chose materials available in the classroom to increase the likelihood that social communication exchanges would maintain and generalize to materials in that environment. That is, if social communication behaviors generalize to other toys, then it is important to ensure that those toys are available in the children's natural environment. However, the issue with selecting stimuli already available in the classroom for generalization probes is that we do not know whether children in this study enjoyed playing with these materials. In addition, the available toys differed considerably more than the toys used for intervention. Mixed generalization results may have been due to the degree of difference in stimuli from intervention to generalization probes. For example, Olin and Olga played with a veterinary set that had one stuffed dog and materials associated with the veterinary profession (e.g., stethoscope, thermometer, ointment, tongue depressor, and syringe); although other play sets were also thematic, they involved just two figurines (e.g., two astronauts) that children gave agency to. Future studies may either select toys for intervention that are already in the classroom and are preferred toys or conduct generalization probes with stimuli that share more of the characteristics of the selected intervention toys.

Multiple-exemplar video instruction did not promote unscripted verbalizations of children with ASD (i.e., response generalization). Response generalization effects were modest across dyads, with the possible exception of Otis, who showed increases above his low and stable baseline. Unlike other studies that found positive results on the use of multiple-exemplar videos for promoting novel responses, the current study shows little to no effects of multiple-exemplar videos on novel responding (Dueñas et al., 2019; MacDonald et al., 2009; Maione & Mirenda, 2006). This may be due to the number of video exemplars each dyad was exposed to. A total of six different videos were made for this study, instead of three video exemplars used in previous studies (Dueñas et al., 2019; MacManus et al., 2015), and children with ASD in this study were exposed to each video two or three times.

The number of video presentations was smaller in this study compared with previous studies. Video exposure may be a factor impacting response generalization as children with ASD may be focused on learning scripts from video before beginning to deviate from scripts.

Social Validity

The study employed two measures of social validity that are not often employed: measuring typical peers' acceptability ratings of the PMI intervention over time and collecting normative comparison data from typically developing peer dyads during play with the same toys. A recent PMI literature review noted only 28% of single-case experimental design and group design studies in their review assessed social validity, and none of the studies assessed social validity of peers (Gunning et al., 2018). Peers in this study provided positive ratings for several intervention features: leaving their classroom for training, watching videos, and playing with their peer with ASD. However, survey results also revealed a change in the peers' perceptions regarding playing with children with ASD. In particular, two typically developing children changed their responses from positive at the beginning of the study to negative in the middle and end of study. Children reported that their peers with ASD were not always sharing toys. This is particularly interesting in light of the aim of the study to reduce adult facilitation. Although adult facilitation may be intrusive during play, adult monitoring may be necessary to ensure positive interactions among children with ASD and their peers (Reiter & Vitani, 2007).

The intensity of the intervention also may have contributed to peers' potential burnout, as sessions were conducted twice a day, every day, for at least 2 weeks. Previous PMI studies have modified peer-training methods by training all members in a classroom rather than dyadic pairs (Laushey & Heflin, 2000). Other studies have spread PMI across the day (Goldstein et al., 1997). These strategies may significantly lessen the feeling of sacrificing one's play time so frequently.

Normative comparison data also highlight an important social validity consideration within PMI. The normative data provide a reference point for evaluating the clinical significance of the intervention. During intervention and maintenance, all participant dyads outperformed the mean social communication exchanges of typical peer dyads, whereas only Olin and Olga performed at normative levels on occasion during baseline. Similar to Chan et al. (2011), we found that the intervention far exceeded the performance of typically developing peers. Although interaction rates may vary considerably depending on several factors, for example, the type of materials, adult supervision, the age of participants, and familiarity with play partners (Goldstein et al., 2007), the effects of PMI have potential benefits in teaching more advanced social communication skills to both typical children as well as children with ASD.

. . . the effects of PMI have potential benefits in teaching more advanced social communication skills to both typical children as well as children with ASD.

Limitations

There are several limitations to this study that may provide direction for future multicomponent PMI research with preschool children with ASD. Despite the increases in social communication exchanges, the lack of balance in initiations and responses by the children with ASD and their peers was disappointing. Future multicomponent PMI studies may need to teach both groups of children to verbally initiate and respond, which could be done in several ways. One approach might involve having children switch roles (being initiator and responder) or exposing both children to video models and self-management strategies, as was done in Laushey and Heflin (2000).

PMI is already a complex intervention; therefore, adding child-direct intervention strategies may have further compounded the complexity of the intervention and affected interpretation of results. It would be beneficial

to empirically test the relative benefit of intervention components within the multicomponent PMI. A component analysis, where individual components of a treatment are systematically applied and assessed for change in behavior, would help identify which components—or combination of components—were responsible for treatment gains (Ward-Horner & Sturmey, 2010).

Unfortunately, the frequency of video-modeling presentations delivered to children with ASD during 30-s lulls in social communication was not monitored. These data would be useful in understanding how long interventionists may need to follow this procedure before fading it completely. In fact, during initial intervention sessions, presentation of video models may have disrupted play sessions, as the presentation of videos took 7 to 10 s.

Several issues concerning IOA should be considered. IOA for unscripted verbalization was below acceptable thresholds for single-case experimental design research, 80% to 90% on average, outlined by WWC (2017); this is a limitation of the study. Although discrepancies among coders were resolved by the first author, this raises potential issues regarding the operational definition and training for coders who are expected to identify unscripted verbalizations in future studies. In addition, the range of IOA during intervention across dependent variables was wide (e.g., 50%–100%); low scores occurred primarily at the beginning of the intervention, which were resolved with calibration. Part of the early discrepancies had to do with becoming familiar with the idiosyncratic speech of participants with ASD (e.g., speech unintelligibility, repeating verbalizations, and word approximations). Future studies should consider more systematic forms of IOA training throughout the study, for example, conducting additional calibration when IOA falls below 80%.

Practical Implications

The current intervention was delivered by the intervention developer and not by practitioners (e.g., speech pathologists or special education teachers). Although the intervention was delivered in the inclusive classroom and teachers were involved in peer selection, material selection, and social validity data obtained for typically developing peers, we know little about the feasibility and acceptability of the intervention in inclusive preschool classrooms when implemented by practitioners. This intervention has multiple components—a peer-training package (self-monitoring using video and reinforcement) and video modeling for focal students—that may impact feasibility in general education settings. However, students with developmental disabilities, including ASD, often require treatment across a wide range of developmental goals and interdisciplinary teams to deliver interventions on those goals (National Institute for Health and Care Excellence, 2013). Therefore, future research should involve teaching, coaching, and supporting practitioners who are part of an interdisciplinary team within an education setting (e.g., speech pathologist, social workers, board-certified behavior analyst) in the implementation of the intervention and soliciting social validity data from practitioners to examine the likelihood that they would adopt the intervention.

Conclusion

Overall, the multicomponent PMI involving the use of self-management and video modeling demonstrates that children with ASD can learn to initiate to typically developing peers during play with minimal adult facilitation. And typically developing peers can learn to respond to their peers reliably with self-management. Consistent with previous investigations evaluating multicomponent PMI, combining interventions that target both children with ASD and their peers may result in improved social communication between children during play. An emerging body of literature has begun to combine PMIs and child-direct intervention strategies to meet the complex social needs of young children with ASD and promote generalization. Direct social instruction delivered to children with ASD and peer training on being responsive

social partners are hypothesized to lead to more robust social communication outcomes (Whalon et al., 2015).

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ORCID iDs

Ana D. Dueñas https://orcid.org/0000-0003-0531-321X

Howard Goldstein https://orcid.org/0000-0003-3270-2739

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